Project 3: E-Commerce Sales Analysis Using SQL

DecodeLabs Data Analytics Internship

📌 What This Project Is About

This project uses SQL to extract actionable business insights from an e-commerce sales dataset. Raw transactional data was queried, cleaned, and analyzed to uncover patterns across revenue, customers, products, marketing channels, and order fulfillment.

The core principle: **Transform raw data into meaningful business decisions.**

🎯 Objectives

- Analyze revenue performance
- Understand customer behavior
- Measure product performance
- Evaluate marketing effectiveness
- Assess order fulfillment performance

📊 Dataset Overview

Property	Detail
Dataset	Orders Table
Total Records	1,200 Orders
Database	MySQL
Database Name	Project_3

Columns in Dataset:

- OrderID, Date, CustomerID, Product, Quantity, UnitPrice

- ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber
- ItemsInCart, CouponCode, ReferralSource, TotalPrice

🛠️ Tools Used

- MySQL (Database Management System)
- SQL (Structured Query Language)

🗄️ Database Setup

```sql
CREATE DATABASE Project_3;
USE Project_3;
```

🔍 Data Cleaning

Issue Found: Missing CouponCode Values

Finding: 309 records had empty CouponCode values

Resolution:

```sql
UPDATE orders_table
SET CouponCode = 'No Coupon Used'
WHERE CouponCode = '';
```

Impact: No records lost — dataset completeness improved ✅

📋 Queries & Business Insights

1. Revenue Analysis

Query	Result	Business Insight
Total Revenue	**$1,264,738**	Business generated over $1.26M in sales
Average Order Value	**$1,053.95**	Customers spend ~$1,054 per transaction

2. Customer Analysis

Query	Result	Business Insight
Unique Customers	**1,189**	Strong customer base acquired

3. Order Fulfillment

Query	Result	Business Insight
Revenue Lost to Returns	**$243,268**	Significant revenue lost post-delivery
Revenue Lost to Cancellations	**$276,390**	Even greater loss before fulfillment
Combined Loss	**$519,658**	41% of revenue at risk

💡 Key Findings

- Total Revenue exceeded **$1.26 million**
- Average Order Value was approximately **$1,054**
- **1,189 unique customers** made purchases
- Returns resulted in **$243,268** in lost revenue
- Cancellations resulted in **$276,390** in lost revenue
- Marketing channels differ significantly in revenue contribution
- Product performance varies across categories
- Coupon usage provides insight into promotion effectiveness

✅ Final Recommendations

1. Increase focus on top-performing products
2. Optimize marketing spend toward high-revenue referral sources
3. Reduce cancellations through improved order processing
4. Investigate return causes to recover lost revenue
5. Expand successful promotional campaigns
6. Develop customer retention strategies
7. Use revenue trends for forecasting and inventory planning

📂 Files in This Repository

File	Description
`project_3_dataset.csv`	Raw dataset imported into MySQL
`project_3_queries.sql`	All 18 SQL queries used in analysis
`project_3_report.docx`	Full written report with findings
`screenshots/`	Query result screenshots
`README.md`	This documentation file

🔢 Full Query List

#	Query	Category
1	View full dataset	Inspection
2	Count total orders	Inspection
3	Preview first 10 records	Inspection
4	Check unique values	Cleaning
5	Count coupon usage & fix nulls	Cleaning
6	Total revenue generated	Revenue

#	Query	Category
7	Average order value	Revenue
8	Total unique customers	Customer
9	Revenue by product	Product
10	Quantity sold by product	Product
11	Revenue by payment method	Payment
12	Revenue by referral source	Marketing
13	Orders by status	Fulfillment
14	Revenue lost through returns	Fulfillment
15	Revenue lost through cancellations	Fulfillment
16	Average cart size by referral source	Behavior
17	Most used coupon code	Promotional
18	Revenue by year	Trend

Author

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